Deep Learning Based Emotion Detection in Voice Data Using Convolutional-LSTM Networks

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***Abstract*—Stress detection is a critical component of mental-health monitoring and support systems. In this study, we present a novel approach for automated stress detection utilizing deep learning techniques, specifically Convolutional Long Short-Term Memory (ConvLSTM) networks. Our model is designed to process voice data, making it suitable for applications in which stress manifests over a sequence of observations. We constructed a neural network architecture that combines convolutional layers to capture temporal patterns with LSTM layers for sequential analysis. The model was trained on a labeled dataset, enabling it to classify stress levels accurately. We employed the categorical cross-entropy loss function and Adam optimizer to optimize the model parameters. Our experimental results demonstrated the effectiveness of the proposed approach for stress detection. We achieved a high accuracy and robustness, making it a valuable tool for real-time stress monitoring and intervention systems. We believe that the combination of ConvLSTM networks and adaptive learning rate techniques can pave the way for more accurate and reliable stress detection systems in various domains, including healthcare and mental health.**

Keywords— Convolutional-LSTM, Stress detection, emotion, optimizer, classifier, deep learning

I. INTRODUCTION

Stress can be also caused by emotional states like despair, worry, rage, grief, guilt, and low self-esteem. This emotional state can be categorized as either positive stress or negative stress known as eustress and distress respectively [1]. Stress has a significant impact on the individual's productivity and is one of the key factors that can lead to dementia and depression [2]. Worldwide, the prevalence of stress-related issues is on the rise, stressing the importance of stress identification and measurement [3]. Self-reports have been the most common way of assessing stress levels over the years [4]. From the submitted response, it was evident that each question had a score assigned and the total score was obtained from all the questions answered [5]. However, there are numerous tools for measuring stress in clinical environments [6–8], which are still largely dependent on the human assessment. Additionally, owing to the cultural taboos, the inhabitants of numerous developing countries are often reluctant to seek help from psychologists and mental health facilities. It would also be easier to develop preventive measures such as informing the public about mental health issues. Today's stress measurement methods, including questionnaires, facial expressions, the frequency of eye blinking, changes in voice, and social media activity, are either subjective or have significant validity concerns.

Contemporary approaches to the problem of emotions' recognition employ numerous methods which is a clear indication of its further development and improvement. The first level of top level split is based on the audio features to separate highly triggered emotions from weakly triggered emotions. After the splits, the cascades are assembled using various classifiers such as Gaussian Mixture Models, Linear Discriminant Analysis, and SVMs.

This method proved to be more efficient than the baseline accuracy rate set for the Support Vector Machine. The current recognition of emotions has been enhanced by the use of deep neural networks. Some examples of the iterations include convolutional networks and recurrent neural networks (RNNs) [15]. The research described in introduced a Deep Neural Network (DNN) that was employed to predict emotions at segment level with the use of probability distributions. The Extreme Learning Machine (ELM), a single-hidden-layer neural network, applies them to the task of emotion recognition at the utterance level. This DNN and ELM stacking architecture improves the results of the SVM-based emotion detection and is beneficial, especially for the cases where there are few training data available.

Moreover, there are Convolutional Neural Network (CNN) [14] based approaches that are also used along with sequence modeling techniques. These methods incorporate spectrogram and Mel-frequency cepstral coefficients (MFCC)[9] into a single feature space. This integration of data from various sources enhanced the model's sensitivity to minor changes in emotions.

II. RELATED WORK

Due to the vast number of algorithms and structures that are developed to address various challenges and contexts, the field of emotion detection is constantly changing. In order to improve the effectiveness and robustness of the emotion detection systems, various methods have been proposed by the researchers ranging from traditional machine learning to deep learning. New approaches for the identification of emotions have been developed in recent studies, each of which presents different approaches to identifying emotions in conversation data.

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Emotion detection is the focus of the research detailed in the study and to this end, a weighted pooling method and a bidirectional Long Short-Term Memory (LSTM) network with an attention mechanism are used. This attention mechanism helps the network to focus on the affective parts of a sentence, enhancing the network's ability to identify affect in the text. Meanwhile, proposes a method based on an RNN-based memory network with a multihop attention mechanism. This method incorporates both self and other's contributions into the memory of the conversation, which enables the model to capture emotional summaries of the situation. This approach offers a more detailed analysis of the emotional state of the conversation by considering these factors [16].

The current approaches for inferring emotions from talks include the Dialogue RNN and its variants, BiDialog RNN and BiDialog RNN with attention. These models are implemented with three different Gated Recurrent Units (GRUs) [17], which take into consideration the overall context of the conversation, including speaker states and emotional states. While the effectiveness of these models has been analyzed in the context of text and tri-modal scenarios involving text, audio, and visual, the performance of these models in the audio modality has not been explored in detail.

These advancements in conversational contexts for emotion identification point to the importance of context-aware methods and the integration of multimodal approaches that incorporate text, audio, and visual cues to better assess emotions in conversational interactions. These are the modalities and their specific roles in the emotion detection models which need further elaboration.

III. SYSTEM METHODOLOGY

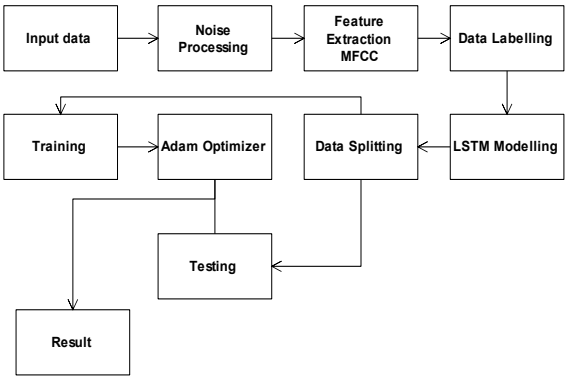

Fig. 1. System Methodology

System design describes a systematic approach to machine learning modeling, especially for sequential or time series data, perhaps aimed at tasks such as speech recognition or signal processing The setup consists of several key steps, each of which contributes enable the entire model development process.

A. Dataset

The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) contains 7,356 files totaling 24.8 terabytes, comprising the entire collection. A total of 24 professional artists (12 women and 12 men) split evenly between genders contributed to this database. These actors use a neutral North American accent and perform two rhyming sentences. The database contains speech and musical expressions that express emotions such as peace, happiness, sadness, anger, fear, shock and contempt. Noise Processing At this stage, the input data is preprocessed to remove noise and improve the quality of the signal. Subsequently, the noise function is important to improve the accuracy of the feature extraction and model training steps.

These data represent recordings or samples of human speech or other acoustic features, which serve as a resource for subsequent analysis and interpretation Music data include detailed information about speech sounds, features surroundings, or compositions. This data is important for tasks, such as speech recognition, emotion recognition, and phonetic segmentation.

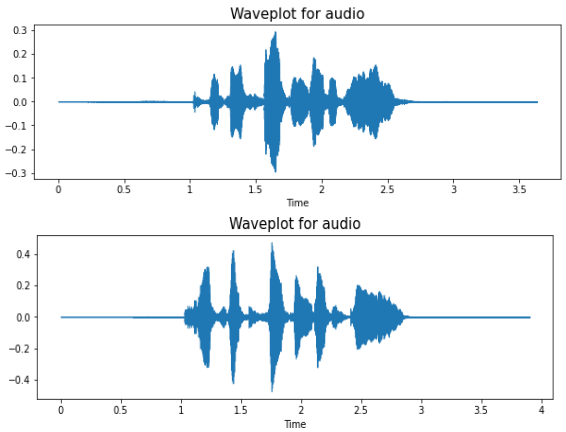

Fig. 2. Wave plot for Stress (Including Anger & Fear)

Practitioners frequently use techniques such as noise manipulation to increase the robustness and flexibility of acoustic data analysis. Noise injection involves deliberately introducing random or controlled noise into the audio signal. This approach serves a variety of purposes, such as simulating real-world scenarios, enhancing training data for machine learning models, testing flexible audio processing algorithms and enabling noise injection analysis of audio signals under various conditions, thus increasing accuracy and reliability -helps towards the results.

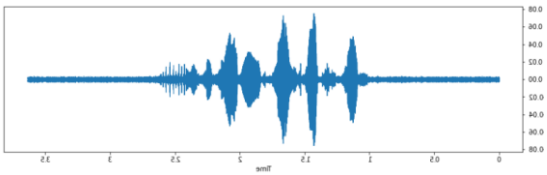

Fig. 3. Noise Injection

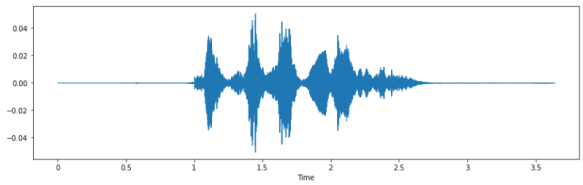

Fig. 4. Pitch of Voice Sample

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Another key operation in audio-signal processing is transfer. Variation includes varying the timing or alignment of audio signal elements or samples. For example, a time shift changes the location in time of sound events in a signal. This function is important for functions such as echo generation, synchronization and alignment of audio data. By manipulating items over time, researchers can create effects, fix convergence problems, or extract specific temporal features from the data. Shifting is a versatile tool in the audio processing toolbox that allows precise control over the temporal quality of a signal.

B. Feature Extraction (MFCC)

In this stage, Mel-Frequency Cepstral Coefficients (MFCC) are extracted from the processed data. MFCC is a widely used technique in audio processing that captures the important characteristics of the signal in a compact form, making it suitable for machine learning applications. The get_features function is a robust method designed for audio feature extraction, enhancing the variability and reliability of the extracted features for machine learning applications such as speech recognition or audio classification. Initially, audio data is loaded using the librosa library, with specified duration and offset parameters to exclude silent sections. A series of random transformations, including noise addition and pitch alteration, are applied to the audio data to introduce variability. This transformation process is performed thrice, each time followed by the extraction of features using the extract_features function. The resulting feature sets, each comprising the first 108 features to maintain consistency, are stored in a list. This multi-step process ensures that different perspectives of the audio data are captured, thus improving the generalization capability of machine learning models trained on these features. The method's design ensures that the models are better equipped to handle real-world audio data variability, leading to more accurate and reliable classification or recognition outcomes.

C. Data Labelling and Splitting

Exclusions are then listed by target class or outcome. Data labeling is important for supervised learning, where the model is trained to map input to corresponding outcomes. The labeled data set is divided into training and test subsets. This step ensures that the model can be tested on unobserved data, which is important for assessing its generalization performance. During the training process, a subset of training is used to fit the model parameters. In this configuration, the ADAM optimizer may be in charge of the training phase.

D. Adam Optimizer

ADAM (Adaptive Moment Estimation) is an optimization algorithm used to iteratively update network weights based on training data. It combines the advantages of two other extensions to stochastic gradient descent, such as the adaptive gradient algorithm (AdaGrad) and the root-mean-square propagation (RMSProp) The choice of the optimizer is established by constructing the ADAM optimizer, a is widely used on the optimization algorithm that during training This optimizer is important for updating model weights and biases in the training process to reduce defined loss functions that optimize learning rate, which is important for accurate stress detection Model accuracy was tracked metrically to measure performance.

To maintain the quality of the training and strictly control the number of courses, a reduction in the number of courses has been introduced. On the Plateau callback, Reduce LR, rlrp, was configured for loss of integrity. When validation loss plateaus for a certain number of epochs (as determined by the "patience" parameter), the learning rate is reduced by a factor of 0.5, thus enabling positive changes in later stages of training This adaptive learning rate mechanism ensure efficient convergence and help avoid local minima in the optimization process.

The training process was automatically set to run for a fixed number of times (100 in this case), during which time the model learned from the training data (x_train and y_train) in groups of size 128. Validation data sets (x_test and y_test) were used to check the performance and general applicability of the model. The history of the training set, including the loss and accuracy metrics of the training and validation sets, was captured for analysis and subsequent analysis

E. LSTM Modelling

The core of the system is the long-term memory network (LSTM), which is a recurrent neural network (RNN) that is well suited for sampling sequential data due to its ability to span long-term reliability so

The neural network architecture incorporates a long-term and short-term memory network (LSTM), optimized for sequential data analysis. Initially, the model applies convolutional operations at each time step independently, facilitated by the TimeDistributed Conv1D layer. Batch normalization ensures robust and fast training. The filtered temporal features are then flattened to match the LSTM layer.

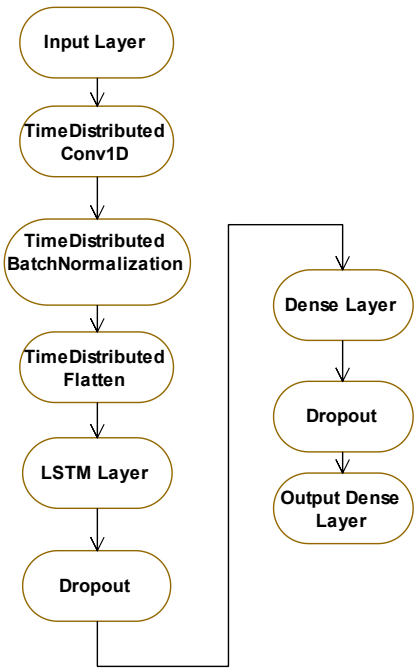

Fig. 5. LSTM flowchart

The basic component, the LSTM layer with 32 units, processes this series of inputs to determine the duration dependence and temporal parameters. The drop-out

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mechanism is set to reduce overload by freely activating a small fraction of muscles during training. After LSTM, a dense layer refines the object again, before the final output layer uses softmax activation to divide the audio data into one of 8 predefined classes This process includes convolutional and iterative layers together well, harnessing their potential for complex audio passages. Model description as below:

```
# Initialize the Sequential model
model = Sequential()
# Add a TimeDistributed Conv1D layer
model.add(TimeDistributed(Conv1D(filters=16,
kernel_size=3, padding='same', activation='relu'),
input_shape=input_shape))
# Add TimeDistributed BatchNormalization
model.add(TimeDistributed(BatchNormalization()))
# Flatten the outputs for each time step
model.add(TimeDistributed(Flatten()))
# Add an LSTM layer to capture temporal dependencies
model.add(LSTM(units=32))
# Add a Dropout layer to prevent overfitting
model.add(Dropout(rate=0.2))
# Add a Dense layer with 32 units and ReLU activation
model.add(Dense(units=32, activation='relu'))
# Add another Dropout layer
model.add(Dropout(rate=0.2))
# Add an output Dense layer with 8 units and softmax
activation
model.add(Dense(units=8, activation='softmax'))
# Print the model summary
model.summary()
```

Once the model is trained, it is evaluated on the testing subset to measure its performance. Testing provides insights into how well the model generalizes to new, unseen data. The final step involves interpreting the model's outputs and results, which could include accuracy, precision, recall, and other relevant metrics. The results are then used to make informed decisions or further refine the model.

IV. RESULT AND ANALYSIS

1) The information presented in the "Results and Analysis" section presents the performance parameters of the classification model, especially for two different groups: "Anger" and "Sadness" These parameters are important for the performance of the model is investigated in the distribution of samples of this sensitivity . Category: This displays the category labels for which performance metrics are reported. In this case, there are two categories: "anger" and "sadness."

2) Accuracy is a metric for measuring how well it predicts patterns of a given class. The accuracy of the "angry" class was 0.82, meaning that if the model correctly classified a sample as "angry" 82% of the time. Similarly, the accuracy of the "sad" group is 0.56, which means that when the model identifies the correct illustration as "sad," it does so 56% of the time.

3) Recall: The capacity of a model to accurately identify instances of a class out of all the actual instances of that class is measured by recall, which is also known as the sensitivity or true positive rate. Recall for the "angry" class is 0.82, meaning that 82% of all actual "angry" instances were correctly identified by the model. The recall for the "sad" class is 0.44, which means that 44% of all actual "sad" instances are correctly identified by the model.

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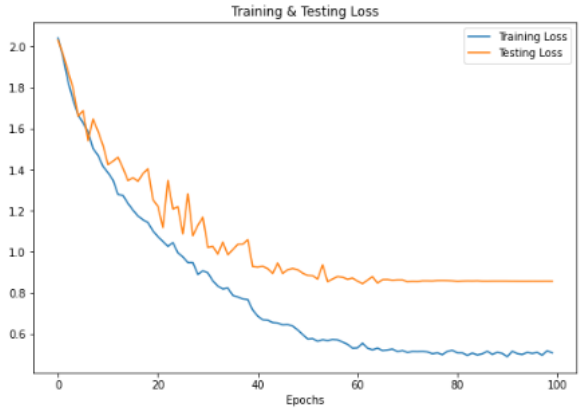

Fig. 6. Training and Testing Loss

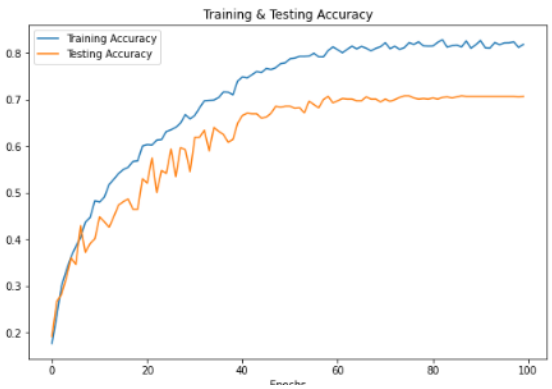

Fig. 7. training & Testing Accuracy

4) F1-Score: The harmonic mean of recall and precision is known as the F1-score. It offers a fair evaluation of the performance of a model, particularly when working with unbalanced datasets. The F1-score for the "angry" class is 0.82, which represents a balanced accuracy score for this class. Owing to the trade-off between precision and recall for the "sad" class, the F1-score was 0.49, which is lower.

5) Support: Support indicates the number of instances of each class in the dataset. There were 111 occurrences of the "angry" class and 108 instances of the "sad" class.

6) Accuracy: The model's overall cross-class accuracy was 0.71, or 71%. This statistic reflects the percentage of the dataset's total number of instances that were successfully categorized.

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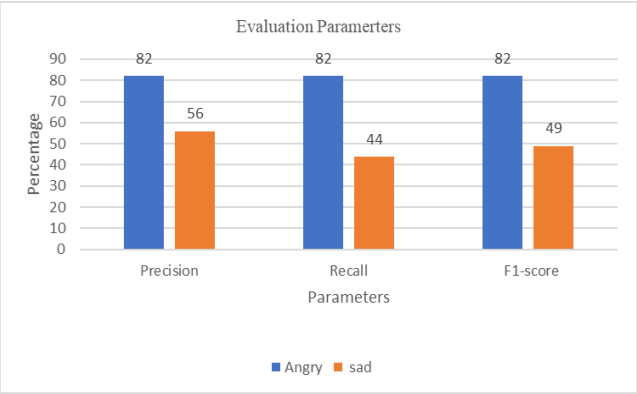

Fig. 8. Precision and Recall for Angry and sad class

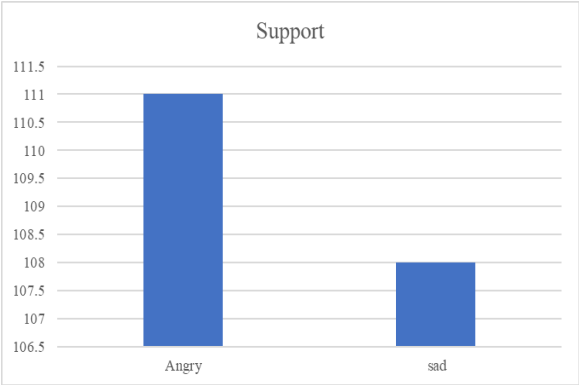

Fig. 9. Support for Angry and Sad class

These performance metrics provide an in-depth evaluation of how well the model is distinguishing between the "angry" and "sad" classes. While the model demonstrates strong precision and recall for the "angry" class, indicating reliable predictions for this emotion, its performance is comparatively weaker for the "sad" class, as reflected by lower precision, recall, and F1-score. An overall accuracy of 71% indicated the general performance of the model across both classes. These metrics are essential for understanding the strengths and weaknesses of the model and can guide further refinement or optimization efforts.

TABLE I. EVALUATION PARAMETERS TABLE

Class	Precision	Recall	F1-score	Support
Angry	0.82	0.82	0.82	111
sad	0.56	0.44	0.49	108

V. CONCLUSION

In conclusion, our research represents a significant step in the stress-detection domain within time-series audio data. Through careful data pre-processing, the employment of Convolutional-LSTM networks, and the judicious use of noise injection and time shifting, we demonstrated the potential for accurate emotion recognition, with a particular focus on distinguishing between anger and sadness. Although our model exhibited commendable precision and recall for anger, there remains room for refinement in recognizing sadness. Nevertheless, our findings underscore the importance of leveraging audio-based emotion analysis in diverse applications, from mental health support to enhancing human-computer interaction. As we conclude this chapter, we anticipate further advancements and applications in the ever-evolving landscape of audio emotion analysis.

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